Solutions and teaching notes, chapter 9

Teacher use. The following answers concern the tasks not solved in the lesson. First have pupils make their approach explicit; provide graduated help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. $2^7 = 128$. *Action:* Reconstruct the product of seven factors.
2. $\frac{1}{9}$. *Action:* Go back to the reciprocal power.
3. $x = 2$. *Action:* Isolate the unknown while maintaining equivalence.
4. $2x + 2$. *Action:* Apply the sum rule for derivatives.
5. $0{,}8$. *Action:* Subtract the rate from 1.
6. 5×2^n. *Action:* Count the n multiplications.

Investigation

The condition $f\prime = f$ gives Delta$y \approx hy$. With h= $\frac{1}{4}$, the values are 1; 1.25; 1.5625; 1.953125. This construction approximates the function and does not replace its characterisation.

Direct applications

1. $y = e + e(x - 1) = ex$.
2. e^{2x+1} ; $x + 2 > 2x$, therefore $x < 2$.
3. $g\prime(t) = -6e^{-2t} < 0$ on $\mathbb{R}$: strictly decreasing.
4. $u_n = 5(e^{0{,}3})^n$; the ratio over two units is $e^{0{,}6}$.
5. Apply the proved inequality with argument 2 x ; equality only at x=0.

Exercises: methods and answers

Exercise 9.1 · Transform. e^2 ; e^{5x} ; $1 + e^{-x}$. For the last expression, distribute and simplify the products.

Exercise 9.2 · Equations and inequalities. Since it is strictly increasing: 2 $x - 1 = x$ + 3, so x=4. Then $-3x \geq x - 2$, i.e. $x \leq \frac{1}{2}$.

Exercise 9.3 · Product and maximum. $f\prime = (1 - x)e^{-x}$; increasing up to 1, then decreasing. Maximum $f(1) = \frac{1}{e}$. Values at the endpoints 0 and 4 e^{-4} lower.

Exercise 9.4 · A modelled population. 200 ; factor $e^{0{,}1}$; rate $e^{0{,}1} - 1 \approx 10{,}52\%$. The resources and the carrying capacity of the environment are not taken into account; it is a model over a limited period. The two required curves pass through (0;1); the first increases, the second decreases. They are symmetric about the vertical axis; see the figure in the section From discrete to continuous.

Exercise 9.5 · Search for a threshold. Test n=0,1,... in turn with a limit on the number of iterations. $Q(4) \approx 10{,}8268 \geq 10$, $Q(5) \approx 6{,}5668 < 10$; decrease proved, so the first integer is 5.

Exercise 9.6 · A symmetry. $f(-x) = f(x)$: even. $f\prime = e^{x} - e^{-x}$ has the sign of x, so minimum 2 at 0. Another proof: $f(x) - 2 = (e^{\frac{x}{2}} - e^{-\frac{x}{2}})^{2}$.

Exercise 9.7 · Uniqueness under an assumption. $g\prime = f\prime e^{-x} - fe^{-x} = 0$. On $\mathbb{R}$, g is constant, and g(0)=1. Thus $f(x) = e^{x}$. This proof of uniqueness uses an exponential already constructed; it does not prove existence.

Exercise 9.8 · Delayed review. $\frac{u_{n+1}}{u_{n}} = e^{2}$; first term e. Sum $e(1+e^{2}+e^{4}) = \frac{e(1-e^{6})}{1-e^{2}}$. Here the direct sum is just as efficient.

CQFD, exercise 18 p. 187, questions a, b and c. a : e^{3x}. b : e^{2}, since e^{x+2} is divided by e^{x}, always non-zero. c: e^{x+1}. Do not confuse the number e with the variable x.

CQFD, exercise 30 p. 187, questions a, b and c. a : $x \leq 0$. b : $2x > 1$, therefore $x > \frac{1}{2}$. c : $-3x \geq 2x + 1$, therefore $x \leq -\frac{1}{5}$. Since the exponential is strictly increasing, the direction is kept; dividing by -5 reverses it in c.

MCQ: answers and reasons

MCQ 1.

1. B. Initial condition.
2. A. Addition property.
3. D. Factor -2.
4. C. Strictly increasing and $e^{0} = 1$.
5. B. Positivity of the exponential.

MCQ 2.

1. C. Compare the exponents.
2. A. Ratio of consecutive terms.
3. D. Value 1 and slope 1.
4. B. Product rule.
5. C. Minimum of the difference.

True or false

1. False: at 0, 1 is different from 2.
2. True: its derivative $-3e^{-3t}$ is negative.
3. False: the discrete rate is $e^{0,1} - 1$, i.e. about 10.52%.

Differentiated teaching. If a pupil is stuck on a technical point, suggest only the first step (domain, factor or reference sample space), then remove the help. For an independent student, ask for a second method and the handling of a boundary case. Orally, allow an initial rephrasing in the language the pupil is fluent in, then require a precise mathematical sentence in French.